Market Survey

Evaluating The Market Capacity Of SUPERCAPACITORS

Sudeshna Das is director at ComConnect Consulting

Although supercapacitor technology is at a nascent stage in India, opportunity for this technology is huge considering the various application sectors. This report explores the potential for supercapacitors in India

Supercapacitors or ultracapacitors are charge storage devices that store electrical charges via electrochemical and electrostatic processes. These have an unusually high energy density as compared to common capacitors. Due to their beneficial properties like fast charging ability, superior low temperature performance, long service and cycle life, and reliability, supercapacitors hold the potential to replace or complement traditional batteries and capacitors in several applications. These are already being used worldwide in a number of applications ranging from automotive, renewable energy to electronics.

Supercapacitors are steadily paving the way for hybrid power storage applications, such as complementary batteries especially in two-wheeler applications. Various market reports estimate the global demand for supercapacitors to grow tremendously, primarily driven by different consumer electronics and automotive applications, to provide backup power. Supercapacitors provide the necessary power backup required for the smooth functioning of applications such as video calling, cameras, wireless communications and GPS navigation.

Other industrial handheld devices—such as GSM/GPRS and radio frequency identification (RFID) communication tools, LED flashlights, thermal printers, barcode scanners and GPS chips—can also be operated more conveniently with the help of supercapacitors, to provide the required power boost. Using supercapacitors in line with batteries in these electronic devices increases the life cycle of conventional batteries, by reducing the load of voltage drops. Thus, battery runtime and operational life is improved extensively by using supercapacitors.

The current practice, across the globe, of upgrading to power generation from renewable resources to reduce rapid depletion of natural resources is also expected to drive the market for supercapacitors in the coming years.

Market scenario

As per IDTechEx report, the global supercapacitor market is expected to reach US$ 8.3 billion by 2025. The market is projected to grow at a compound annual growth rate (CAGR) of 30 per cent till 2025 (Fig. 1).

Consumer electronics and automotive will be the highest revenue-generating segments during this period. Although the supercapacitor market is at a nascent stage, ample emerging growth opportunities are available for the future. Highlighting features such as regenerative braking and easy application in hybrid electric vehicles (EVs) are making supercapacitors the best suited device in automotive applications. Increasing demand for renewable energy generation has been observed in countries across Europe, Asia and USA, which would further fuel the supercapacitor market growth.

As per TechSci Research, the Indian supercapacitors market is projected to grow at a CAGR of around 16 per cent during 2017 - 2022 on account of huge demand for supercapacitors from the consumer electronics segment. Supercapacitors are used in several devices in consumer electronics category including smartphones, laptops, TVs, cameras, lighting appliances and GPS devices.

Moreover, evolution of supercapacitors as a sustainable energy storage solution, growth of EVs market and increasing capacities of supercapacitors resulting in their applications in wind and solar power sectors is anticipated to boost their demand in India in the next five years.

According to 'Supercapacitor Market Landscape Study' prepared by Electronic Industries Association of India (ELCINA), an

Global supercapacitor market 2014-2025 (Source: IDTechEx report)